Ariel Gjaci

Postdoctoral Researcher | Multimodal Generative Learning | Spatial Audio | Human Motion

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Website | LinkedIn | GitHub | Google Scholar

Postdoctoral researcher in artificial intelligence and robotics, with a background in biomedical engineering. Research spans multimodal learning, generative models, human motion, and spatial audio, from dataset creation and model design to distributed training and evaluation.

PROFESSIONAL EXPERIENCE

ITALIAN INSTITUTE OF TECHNOLOGY (IIT) | Postdoctoral Researcher

Feb 2025 - Present | Genoa, Italy

Multimodal learning and spatial audio for immersive environments in the European XTREME project.

- Developing few-shot room impulse response generation from sparse recordings, source/receiver geometry, and depth images, using counterfactual decay interventions to learn shared reverberation control alongside query-dependent acoustic memory (ongoing research).
- Built Eigenmike 64 processing tools for MVDR, DAS, and MaxRE beamforming, binaural audio, and acoustic energy mapping on 3D room point clouds.
- Implemented audio/video synchronization and 360-degree acoustic overlays for spatial-audio recordings.

UNIVERSITY OF GENOA | PhD Researcher

Nov 2021 - May 2025 | Genoa, Italy / London, UK

Research on multimodal human motion generation and domain generalization, including a research collaboration with King's College London.

- Designed ALaDiT, a multimodal diffusion-transformer for co-speech motion generation, within SICAGE (ECCV 2026, main conference).
- Built and released TED4C-L: 106.45 hours of synchronized motion, audio, and text features from 764 speakers across four cultural groups.
- Learned speaker-independent cultural representations with Fishr and adversarial domain generalization, evaluated on speakers excluded from training.
- Developed semantic-similarity algorithms for symbolic and deictic gesture labeling (IEEE RO-MAN 2024).
- Analyzed cultural cues in interaction and affective data using classical and deep learning models (AIRO 2022; AHRI at ACII 2024).

AKKA TECHNOLOGIES | Software Engineer

May 2021 - Oct 2021 | Italy

- Developed Python backend software for aircraft panels and presented a working demo to stakeholders.

RESEARCH FOCUS AND TECHNICAL SKILLS

Research Focus: 3D audio-visual learning; Human motion generation; Multimodal generative learning; Domain generalization.

Programming: Python; C/C++; R; MATLAB/Simulink; LaTeX.

ML Frameworks: PyTorch; Scikit-learn; TensorFlow.

Systems: Linux; ROS 1; Windows.

Tools: Git; SLURM.

Model Training: Distributed multi-GPU training.

EDUCATION

UNIVERSITY OF GENOA | Ph.D. in Artificial Intelligence and Robotics

Nov 2021 - May 2025

Research collaboration with King's College London; thesis defended in May 2025.

UNIVERSITY OF GENOA | M.Sc. in Robotics Engineering

2018 - 2021 | 103/110

Thesis: Culture-Aware Co-Speech Gestures Using Generative Adversarial Networks.

Developed speech-driven GAN gesture generation tested on the Pepper robot (AIRO 2021; International Journal of Social Robotics 2022).

UNIVERSITY OF GENOA | B.Sc. in Biomedical Engineering

2014 - 2018 | 92/110

Thesis: Characterization of the Activity Evoked by Stimuli in Hippocampal Neuronal Networks.

PUBLICATIONS

A. Gjaci, A. Sgorbissa, V. Murino. SICAGE: Speaker-Independent Culture-Aware Gesture Generation using TED4C-L Dataset. European Conference on Computer Vision (ECCV), main conference, pp. 392-411. 2026.

A. Gjaci, V. Schmuck, A. Sgorbissa, O. Celiktutan. Exploring Cultural Cues in Multimodal Interactions. AHRI Workshop at ACII, pp. 306-313. 2024.

A. Gjaci, C. T. Recchiuto, A. Sgorbissa. Labeling Sentences with Symbolic and Deictic Gestures via Semantic Similarity. IEEE RO-MAN, pp. 477-484. 2024.

A. Gjaci, C. T. Recchiuto, A. Sgorbissa. Towards Culture-Aware Co-Speech Gestures for Social Robots. International Journal of Social Robotics, 14(6), pp. 1493-1506. 2022.

A. Gjaci, L. Oneto, C. T. Recchiuto, A. Sgorbissa. Culture Awareness in Intelligent Systems. AIRO Workshop. 2022.

A. Gjaci, C. T. Recchiuto, A. Sgorbissa. A GAN-based Approach for Generating Culture-Aware Co-Speech Gestures. AIRO Workshop. 2021.

TEACHING AND SERVICE

- Teaching assistant: Foundations of Informatics (C++), 60 hours; Mobile Programming (Android/Kotlin), 30 hours.
- Co-organizer of the ECCV workshop Expressive Encounters.
- Reviewer for AAAI, ECCV, ACM Multimedia, IROS, ICMI, and the International Journal of Social Robotics.
- Session chair at IEEE RO-MAN 2024.

TRAINING

Topics in Modern Machine Learning (MaLGa, 2023); Robotics and Intelligent Machines Summer School (Scuola Superiore Sant'Anna, 2023); Coursera courses in deep learning and NLP (2021-2022).

LANGUAGES

Italian (native); Albanian (native); English (fluent).